Revisiting Communication in Federated Learning: A Comparative Study of Parameter and Knowledge-Based Strategies

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Abstract. Federated Learning (FL) enables collaborative model training without sharing raw data, making it particularly suitable for privacy-sensitive healthcare applications. However, the effectiveness of FL strongly depends on the communication mechanism used between clients and the server, especially under heterogeneous and non-IID data distributions.
In this work, we present a systematic experimental comparison of parameter-based and knowledge-based communication strategies in FL, including Federated Averaging (FedAvg), knowledge distillation with weight sharing, soft-label communication, and prototype-based methods. Experiments are conducted on three wearable fall detection datasets under a realistic client-disjoint setting.
Results show that FedAvg achieves the best global performance under non-IID data. However, knowledge-only strategies achieve more competitive local performance, suggesting their relevance for personalization-oriented federated learning.

Keywords: Federated Learning · Non-IID Data · Model Communication · Knowledge Transfer · Healthcare AI

1 Introduction

The rapid adoption of wearable devices and Internet of Medical Things systems has enabled continuous health monitoring and data-driven medical applications [7]. However, the sensitive nature of healthcare data poses significant challenges to traditional centralized machine learning approaches, which require aggregating data in a single location and increase the risk of privacy breaches.

Federated Learning has been proposed as a privacy-aware alternative, allowing multiple clients to collaboratively train models without sharing raw data [8] [14]. In this paradigm, only model updates are exchanged, making FL particularly suitable for distributed healthcare scenarios such as wearable-based monitoring.

Despite its advantages, FL remains highly sensitive to the choice of communication mechanism. The widely used Federated Averaging (FedAvg) algorithm aggregates local model parameters but suffers from performance degradation in the presence of heterogeneous and non-IID data [10]. Additionally, sharing gradients or model parameters can still expose sensitive information through inference and reconstruction attacks [12].

These limitations motivate the exploration of alternative communication strategies that can improve robustness while reducing potential privacy leakage. In this work, we experimentally evaluate different communication mechanisms for federated learning, with a particular focus on Knowledge Distillation-based approaches as an alternative to traditional weight aggregation.

The evaluation is conducted in the context of fall detection using wearable sensor data, a representative healthcare application involving distributed and privacy-sensitive data.

Wearable-based fall detection is typically formulated as a binary classification or anomaly detection problem, where models must distinguish rare fall events from common daily activities under highly imbalanced and user-specific data distributions [4] [15].

2 Contributions

The main contributions of this work are:

- A systematic experimental comparison of communication mechanisms in federated learning, including parameter aggregation and knowledge-based strategies;
- An evaluation of these communication strategies under realistic non-IID data distributions derived from multiple wearable datasets;
- An analysis of the performance trade-offs between global accuracy and communication design, highlighting the impact of different information exchange mechanisms;
- Empirical evidence that knowledge-only communication strategies underperform as global models but remain competitive for local client models, making them relevant for personalization-oriented federated learning.

3 Related Work

Federated Learning (FL) has become a widely adopted paradigm for privacy-preserving machine learning, particularly in sensitive domains such as healthcare [8] [14]. The most commonly used approach, Federated Averaging (FedAvg), aggregates locally trained models by computing a weighted average of client parameters. While simple and communication-efficient, FedAvg relies on the assumption that client data distributions are similar. In practice, however, federated settings are characterized by heterogeneous and non-IID data, which can significantly degrade performance and hinder convergence [10] [11].

In addition to statistical heterogeneity, privacy risks remain an important concern in FL. Although raw data are not shared, gradients and model parameters can still leak sensitive information. Prior work has demonstrated that it is possible to reconstruct training samples from shared updates using gradient inversion techniques, raising concerns about the effectiveness of standard FL communication protocols in privacy-sensitive applications [12] [3].

To address these limitations, several alternative communication strategies have been proposed, among which Knowledge Distillation (KD) has gained increasing attention. In KD-based federated learning, clients exchange model outputs or intermediate knowledge instead of full model parameters, enabling collaboration while reducing communication overhead and relaxing architectural constraints [9] [19]. A prominent line of work focuses on logit-based distillation, where soft labels are shared across clients or through a central server, capturing richer information than hard labels and improving robustness under heterogeneous data distributions [9] [6].

More recent approaches extend this paradigm by exploring more compact and abstract forms of knowledge exchange. In particular, prototype-based methods communicate class-level representations, typically computed as feature centroids, to align models across clients while significantly reducing communication cost [17]. These approaches have shown improved robustness in highly non-IID settings and are especially suitable for distributed sensing applications, where data variability across users is pronounced.

Despite these advances, a systematic comparison of different communication mechanisms—particularly between traditional parameter aggregation and KD-based strategies—remains limited in application-specific scenarios such as wearable-based fall detection. In particular, the relative effectiveness of soft label exchange and prototype-based communication under realistic data heterogeneity conditions is not yet fully understood.

Although federated learning has been successfully applied in multiple healthcare domains, its application to wearable-based fall detection remains relatively limited. Existing works demonstrate that FL can improve generalization across distributed users while preserving data privacy, but also highlight challenges such as class imbalance, device heterogeneity, and limited real-world validation [5] [18] [1].

To address this gap, we conduct a systematic experimental evaluation comparing weight-based aggregation (FedAvg) with multiple KD-based communication strategies, including soft-label distillation and prototype-based methods, in a wearable fall detection setting.

4 Methodology

4.1 Federated Learning Setup

We adopt a client–server federated learning architecture implemented using the Flower framework [2]. The federated training stage uses 69 training clients, each corresponding to a local data partition derived from the KFall, SisFall, and UP-Fall datasets, simulating a realistic heterogeneous and non-IID environment [20] [16] [13]. This non-IID setting arises from subject-wise data partitioning, where each client exhibits distinct activity patterns, sensor dynamics, and class distributions.

Training follows a standard round-based procedure. At each communication round, the server distributes a global signal shared across clients. For weight-sharing strategies, this corresponds to the global model parameters. For knowledge-based approaches without weight sharing, the server instead distributes aggregated knowledge signals, such as soft labels or class prototypes.

To ensure a fair comparison, all experiments use the same model architecture, datasets, preprocessing pipeline, and training configuration. The main varying component is the communication mechanism, while the dataset, preprocessing pipeline, base architecture, and training budget are kept fixed.

A multilayer perceptron (MLP) with an input dimensionality of 32 features, hidden layers $[256, 128, 64]$, and ReLU activations is employed as the base model in all experiments.

4.2 Communication Strategies

This section describes the communication mechanisms evaluated in this work. Table 1 summarizes their main characteristics.

Table 1. Comparison of communication strategies evaluated in this work.

Strategy	**Communicated Information**	**Key Properties**
FedAvg	Model weights	Full parameter sharing; strongest global performance; higher privacy risk
KD with weight sharing	Model weights	Parameter sharing with local teacher–student distillation
Soft-label communication	Aggregated soft-label statistics	No weight sharing; compact communication; summarizes client-level predictive behaviour
Prototype-based communication	Class prototypes	Compact representation; reduced communication; limited expressiveness

FedAvg FedAvg is used as the baseline communication strategy. In this setting, each client trains a local model using its private data and sends the updated model parameters to the server. The server aggregates the received parameters by computing their average to obtain an updated global model [14].

In our implementation, we consider an unweighted aggregation scheme, where each client contributes equally to the global update, regardless of its local dataset size. This choice was motivated by preliminary experiments showing that equal weighting improves robustness under heterogeneous data distributions in multi-dataset settings.

KD with Weight Sharing This variant extends the standard FL setup by incorporating knowledge distillation [9] [19] at the client level while still sharing model parameters. Each client trains its local model using a composite loss that combines the supervised classification loss with a distillation term based on the predictions of the global model received from the server (teacher model).

After local training, the updated local model parameters are sent to the server for aggregation. This approach aims to improve generalization by leveraging softened outputs while preserving compatibility with parameter aggregation.

Soft-label Communication Without Weight Sharing In this approach, clients do not exchange model parameters. Instead, each client computes aggregated soft-label statistics from its local model predictions and sends them to the server. The server aggregates these client-level statistics and redistributes the resulting global soft-label signal in the following round. Each client then uses this global signal as an additional distillation target during local training.

This strategy eliminates parameter sharing and reduces communication cost, but the exchanged information is highly compressed and may not capture the full diversity of local prediction distributions.

Class-prototype Communication Without Weight Sharing In the prototype-based approach, clients communicate compact class-level representations instead of model parameters. Each client computes class prototypes by averaging feature embeddings for each class in its local dataset [17].

These prototypes are sent to the server, which aggregates them across clients to obtain global class representations. The aggregated prototypes are then broadcast back to clients and used to align local feature spaces during training.

This method significantly reduces communication overhead and avoids sharing model parameters, but may lose fine-grained information contained in full model updates.

Table 2. Approximate client-to-server communication payload per round.

Strategy	**Information sent by each client**	**Scalars**
FedAvg	Full MLP model parameters	49,665
KD with weight sharing	Full MLP model parameters	49,665
Soft-label communication	Mean predicted fall probability and sample count	2
Class-prototype communication	Positive and negative 64-dimensional hidden prototypes plus counts	132

4.3 Training Procedure

For the main repeated-run comparison, experiments were conducted under a common training budget to ensure a fair comparison between communication strategies.

All experiments are conducted using the MLP architecture described in Section 4.1. During training, a dropout rate of 0.3 is applied to improve generalization.

For KD-based training in the main comparison, the distillation temperature is set to 2.0 and the supervised loss weight α is set to 0.5. In the local-first tuning analysis, these parameters are varied together with dropout and learning rate.

Training is performed over 25 communication rounds with full client participation. At each round, every client trains locally for 100 epochs before sending updates or knowledge representations to the server, depending on the communication strategy.

To address class imbalance at the client level, local training uses a weighted binary cross-entropy loss, where the positive class weight is computed from the ratio between negative and positive samples in each client's local training data.

During local training, the best local model state is selected according to F1-score on the client's local validation split.

For the main strategy comparison, each configuration is evaluated over 10 independent runs using different random seeds, and results are reported as mean and standard deviation across runs. Local-first tuning experiments are reported separately using selected configurations.

5 Experimental Setup

This section describes the experimental setup used to evaluate the proposed communication strategies, including datasets, preprocessing, and evaluation protocol.

5.1 Datasets

The experiments are conducted using three public wearable fall detection datasets: KFall, SisFall, and UP-Fall [20] [16] [13]. These datasets differ in sensing configurations, subject populations, and activity distributions, providing a realistic heterogeneous environment for federated learning. They also exhibit varying levels of class imbalance and demographic diversity, which contribute to realistic data heterogeneity across clients.

Each client corresponds to a subject-specific partition. The federated training stage uses 69 clients across the three datasets.

Table 3 reports the client-disjoint split used in the experiments. The split is performed at subject level, ensuring that no individual appears in more than one split.

Table 3. Client-disjoint train/dev/test split by dataset.

Dataset	Train	Dev	Test	Total
KFall	26	3	3	32
SisFall	30	4	4	38
UP-Fall	13	2	2	17
Total used	69	9	9	87

5.2 Data Preprocessing

To ensure consistency across datasets, all recordings are converted into a common representation prior to training. Accelerometer and gyroscope signals are mapped to six standardized axes: acc_x, acc_y, acc_z, $gyro_x$, $gyro_y$, and $gyro_z$.

Signals are filtered using a fourth-order low-pass Butterworth filter with a cutoff frequency of 5 Hz and resampled to 20 Hz. The processed signals are segmented into 1-second windows with 50% overlap.

The final preprocessing configuration, denoted as *magnitude_features*, augments the six base signals with two magnitude-based features, acc_{mag} and $gyro_{mag}$. Each trial is normalized using z-score normalization before window-level feature extraction.

For each of the eight signals, four statistical features are extracted: mean, standard deviation, maximum absolute value, and signal energy, resulting in a total of 32 input features per window. The classification task is formulated as a binary problem distinguishing fall from non-fall activities.

After window-level feature extraction, missing values are imputed using the median computed from the training split, and features are standardized using a scaler fitted only on the training data. The same transformation is then applied to development and test splits.

This preprocessing pipeline standardizes signals across datasets while preserving relevant motion characteristics, enabling a fair comparison between communication strategies. The use of statistical features provides a compact representation of time-series data, reducing model complexity while retaining discriminative information for fall detection.

Furthermore, the integration of multiple datasets with different characteristics contributes to a realistic non-IID setting, where variations in sensor configuration, subject behaviour, and class distribution naturally emerge across clients.

5.3 Evaluation Protocol

All experiments follow a client-disjoint train/dev/test protocol. Clients are constructed using a subject-wise partitioning strategy, where each client corresponds to one individual from one of the datasets. The client assignment is fixed across all communication strategies, ensuring that no subject appears in more than one split.

The federated training split contains 69 clients, while 9 clients are reserved for development and 9 clients are held out for final testing. This results in a total of 87 clients across all datasets.

Within each training client, local trials are further divided into local training and validation subsets for client-side monitoring. This local validation is distinct from the global development split, which contains held-out clients.

This protocol evaluates generalization to unseen users, which is more realistic for wearable fall detection than a trial-disjoint setup, where data from the same subject may appear in both training and testing. It also induces a naturally non-IID setting, since clients differ in activity patterns, sensor dynamics, dataset origin, and class distribution.

Training is performed over 25 communication rounds with full participation of training clients. At each round, every client trains locally for 100 epochs before sending updates or knowledge representations to the server, depending on the communication strategy.

For the main strategy comparison, each configuration is evaluated over 10 independent runs using different random seeds, and results are reported as mean and standard deviation across runs. Local-first fine-tuning experiments are reported separately using selected configurations.

Model performance is assessed using PR-AUC and F1-score as the primary evaluation metrics. PR-AUC is emphasized due to the inherent class imbalance of fall detection tasks, where fall events represent the minority class. Threshold-based metrics are computed using a fixed probability threshold of 0.5, while PR-AUC and ROC-AUC are computed from predicted probabilities.

During training, only the 69 training clients participate in the federated optimization process, while the dev and test clients remain completely held out.

For weight-sharing strategies, the server produces a final global model, which is directly evaluated on the held-out dev and test clients.

For knowledge-only strategies, no global parameter model is produced. Instead, evaluation is performed by applying the final client models to the held-out dev and test datasets, and averaging their predictions to obtain global metrics.

6 Results

Table 4. Global and local performance of the evaluated communication strategies. Local metrics correspond to client-side validation performance on training clients. Results are reported as mean ± standard deviation over 10 runs.

Strategy	**Global PR-AUC**	**Global F1**	**Local PR-AUC**	**Local F1**
FedAvg	0.9692 ± 0.0024	0.9184 ± 0.0027	0.9153 ± 0.0107	0.7674 ± 0.0053
KD with weight sharing	0.9573 ± 0.0022	0.8661 ± 0.0083	0.9143 ± 0.0065	0.7517 ± 0.0051
Soft-label communication	0.7320 ± 0.0055	0.6016 ± 0.0074	0.8085 ± 0.0062	0.7663 ± 0.0085
Prototype-based communication	0.7012 ± 0.0066	0.6099 ± 0.0093	0.7923 ± 0.0060	0.7505 ± 0.0059

The following fine-tuning analysis is separate from the repeated-run comparison in Table 4; it reports selected local-first configurations optimized for local performance.

Table 5. Effect of local-first tuning for knowledge-only communication strategies. Results correspond to selected configurations and are not repeated-run averages.

Strategy	**Global PR-AUC**	**Global F1**	**Local PR-AUC**	**Local F1**
Soft labels, baseline	0.7320	0.6016	0.8085	0.7663
Soft labels, fine-tuned	0.7374	0.6466	0.8198	0.7864
Class prototypes, baseline	0.7012	0.6099	0.7923	0.7505
Class prototypes, fine-tuned	0.6918	0.6062	0.8154	0.7479

Table 4 summarizes both the global and local performance of the evaluated communication strategies.

FedAvg achieves the best global performance, reaching a PR-AUC of 0.9692 ± 0.0024 and an F1-score of 0.9184 ± 0.0027, confirming its effectiveness in learning a shared global model under heterogeneous conditions. KD with weight sharing achieves the second-best global performance, with a PR-AUC of 0.9573 ± 0.0022.

In contrast, knowledge-only communication strategies, including soft-label and prototype-based approaches, show substantially lower held-out test performance. Soft-label communication achieves a global PR-AUC of 0.7320 ± 0.0055, while prototype-based communication reaches 0.7012 ± 0.0066.

However, a different picture emerges when considering client-level performance. Although these methods underperform in terms of held-out test performance, their client-specific models are substantially more competitive. Soft-label communication achieves a local PR-AUC of 0.8085 ± 0.0062 and a local F1-score of 0.7663 ± 0.0085, approaching the local F1-score of FedAvg. Prototype-based communication reaches a local PR-AUC of 0.7923 ± 0.0060.

These results indicate that knowledge-only strategies are less suitable for learning a single global model, but remain competitive in personalized federated learning scenarios.

The local-first fine-tuning configurations were selected through a lightweight search using 3 communication rounds and 20 local epochs, prioritizing local PR-AUC, followed by global PR-AUC and F1-score.

Table 5 presents an additional local-first fine-tuning experiment for the knowledge-only strategies. For soft-label communication, prioritizing local performance improves both local and global metrics, increasing local PR-AUC from 0.8085 to 0.8198 and local F1-score from 0.7663 to 0.7864. For prototype-based communication, fine tuning improves local PR-AUC from 0.7923 to 0.8154, although global performance slightly decreases.

These results reinforce that knowledge-only communication strategies are better suited for personalization-oriented federated learning settings rather than for global model optimization.

7 Discussion

This study provides empirical insights into how the nature of exchanged information influences federated learning performance under realistic conditions. The results consistently reveal a trade-off between predictive performance and the characteristics of the communication mechanism.

FedAvg achieves the strongest overall performance, confirming that direct parameter aggregation remains highly effective even in heterogeneous multi-dataset environments. The strong performance of the unweighted aggregation variant suggests that treating clients equally can help mitigate biases introduced by imbalanced local datasets.

KD with weight sharing demonstrates competitive PR-AUC but consistently lower F1-score, indicating that while distillation can preserve ranking quality, it does not necessarily translate into improved classification performance in this scenario.

In contrast, communication strategies that avoid model parameter exchange show substantially lower held-out test performance. This suggests that the information contained in soft labels or class prototypes is not sufficient to fully capture the complexity of the underlying data distribution when compared to full parameter sharing.

However, these approaches offer important advantages. By avoiding the exchange of model parameters, they reduce communication requirements and may limit exposure to certain types of inference attacks. Moreover, their relatively stronger performance at the client level indicates that they are better suited for personalized federated learning settings, where the objective shifts from learning a single global model to improving individual client models.

One limitation of the soft-label strategy is that the communicated signal is highly compressed, as it summarizes client predictions rather than transmitting detailed sample-level information or model parameters. Similarly, prototype-based communication relies on compact class-level representations, which may not fully capture intra-class variability across heterogeneous clients.

The fine-grained analysis further highlights the distinction between global and local objectives. While FedAvg remains the most effective approach for learning a shared global model, soft-label and prototype-based communication become more relevant when the objective is to optimize client-specific performance without sharing parameters.

In particular, the locally tuned soft-label configuration improves both local PR-AUC and F1-score, suggesting that increasing the contribution of supervised local training helps preserve client-specific discriminative information while still benefiting from shared knowledge signals. Prototype-based communication exhibits a different behavior: local ranking performance improves, but global performance slightly degrades, indicating that compact class-level alignment may be insufficient to support a strong global decision boundary.

Taken together, these results highlight that communication efficiency and personalization capability are intrinsically linked, often at the expense of global aggregation performance.

8 Conclusion

This paper presented a comprehensive and controlled comparison of communication mechanisms in federated learning, explicitly examining the trade-offs between parameter-based and knowledge-based strategies under realistic non-IID conditions. Through extensive experiments on multiple wearable sensing datasets, we provide clear empirical evidence that the nature of exchanged information plays a critical role in determining both global and local performance.

Our results show that parameter aggregation, as instantiated by FedAvg, remains the most effective approach for learning a high-quality global model, consistently outperforming all alternative strategies. This reinforces the view that full model updates capture richer and more expressive information than compressed communication signals, even in highly heterogeneous environments.

At the same time, we demonstrate that knowledge-based communication mechanisms—despite their substantially weaker global performance—achieve competitive results at the client level. This finding is particularly significant, as it challenges the conventional focus on global model quality and highlights the importance of considering client-specific objectives in federated settings. In this context, compact knowledge representations such as soft labels and class prototypes emerge as viable alternatives when personalization, communication efficiency, or reduced parameter exposure are primary concerns.

Taken together, our findings reveal a fundamental tension between global optimization and communication efficiency in federated learning. Rather than a single universally optimal strategy, the choice of communication mechanism should be explicitly aligned with the target deployment scenario—whether prioritizing a shared global model or individualized client performance.

Building on these insights, future work will explore hybrid communication frameworks that dynamically combine parameter sharing and knowledge transfer, as well as adaptive strategies that tailor communication to the level of data heterogeneity across clients. Additionally, integrating stronger privacy guarantees and evaluating these methods in real-world deployments remain critical directions for translating federated learning into practical healthcare applications.

Acknowledgments: This research was partially supported by national funds through Fundação para a Ciência e a Tecnologia, I.P. (FCT) ISTAR-Iscte Pluriannual Projects UID/04466/2025 (https://doi.org/10.54499/UID/04466/2025) and Project No. 2024.07426.IACDC/2024, supported by measure "RE-C05-i08.M04"

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